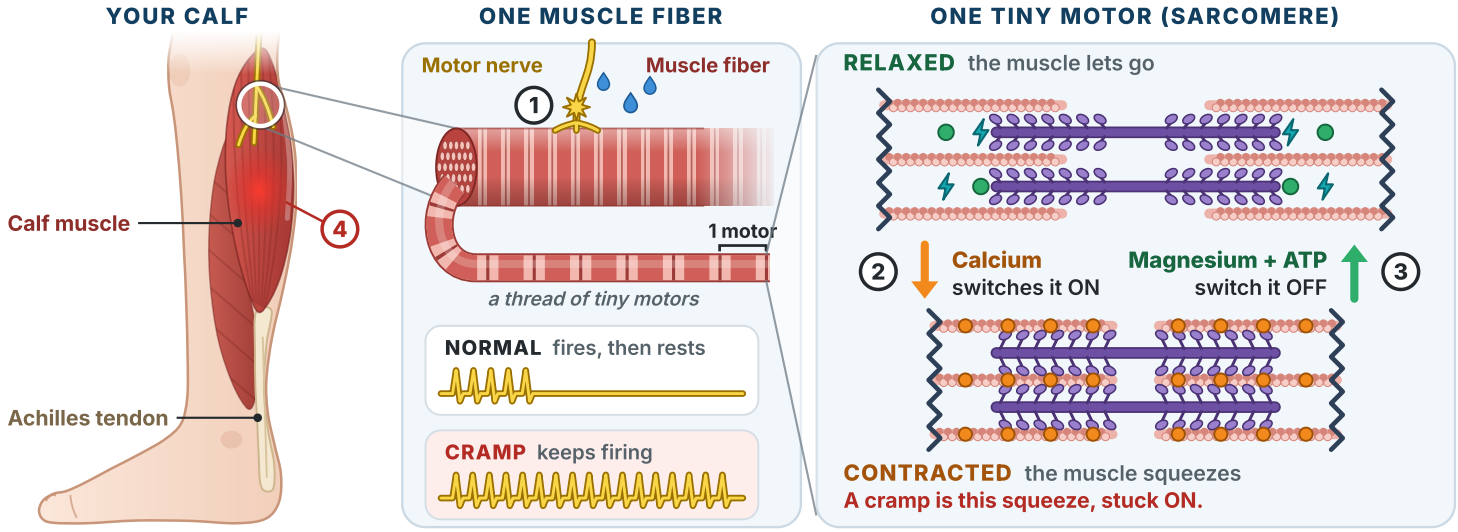
Patient name _____ Date _____

Muscle Cramps and Spasms

Why a muscle locks up, and how to help it let go

A **cramp**, or spasm, is a muscle that suddenly tightens on its own and will not let go. It usually grabs the **calf**, foot or thigh, often at night. Cramps are very common: **up to 6 in 10 adults** have had a night leg cramp, and nearly **half of people over 60** get them. Most are harmless, and simple habits help. Here is what happens inside the muscle.



KEY

Nerve signal · Thin filament (actin) · Calcium: switch ON · Thick filament (myosin) · Magnesium: switch OFF · Z-line: the anchor · Sodium, potassium · Cramp knot · ATP: energy

1. **The nerve says "squeeze."** Sodium and potassium carry the signal into the muscle.
2. **Calcium switches it ON.** Thin and thick filaments grab and slide, so the muscle shortens.
3. **Magnesium and ATP switch it OFF.** The filaments let go and calcium goes back into storage.
4. **A cramp is the switch stuck ON.** A tired or over-excited nerve keeps firing.

WHAT MAKES CRAMPS MORE LIKELY

- Tired, overused muscles
- Sweat, heat, too little fluid
- Sitting still for hours
- Getting older
- Pregnancy
- Water pills and statins
- Low magnesium or potassium
- Diabetes, nerve, thyroid issues

WHAT THE RESEARCH SAYS

HELPED **Stretching at bedtime.** Adults over 55 who stretched calves and hamstrings nightly for 6 weeks had fewer, milder night cramps.

PROMISING **Vitamin K2 (MK-7).** Adults 65+ taking 180 mcg nightly for 8 weeks went from 2.6 cramps a week to about 1; the placebo group rose to 3.6. One trial so far.

MIXED **Magnesium.** Unlikely to prevent ordinary night cramps in older adults (Cochrane review). A small trial tied it to less soreness after hard exercise.

MIXED **CoQ10 with a statin.** One review of trials found less muscle pain and cramping; another found no clear benefit.

Hallegraeff 2012 · Tan 2024 · Cochrane 2020 · Reno 2022 · Qu 2018 · Banach 2015

⚠ GET CHECKED

- A sore calf that is **swollen, warm or red**. **It can be a blood clot. Get care today.**
- On a statin, with **muscle pain or weakness you can't explain**, or **dark urine**. **Call your prescriber right away.**
- Cramps that come **often**, wake you most nights, or come with **weakness or numbness**. **Have a doctor check them.**

Quinine pills are no longer advised for leg cramps.

Lifestyle Recommendations

Stretch before bed
Gently stretch your calves and the backs of your thighs every night before sleep.

Drink, plus salts
Sip water all day. After long, sweaty workouts, add electrolytes, not just water.

Eat your minerals
Greens, beans, nuts and seeds for magnesium. Potatoes and bananas for potassium.

Build up slowly
Add exercise a little at a time. Tired muscles cramp more easily.

Stop a cramp
Straighten the leg, pull your toes toward you, rub the knot, add warmth.

Check medicines
Water pills, statins and some inhalers can play a part. Ask first.

Nutrient Solutions

What your body needs, and the NutriDyn formula that delivers it

Help muscles let go
magnesium + calcium, theanine, passion flower

START HERE



Muscle Eze
90 capsules · 30 servings

- Helps relax muscle tissue
- Supports calm before sleep

IN EACH SERVING
Per 3 capsules: 200 mg magnesium (citrate), 100 mg calcium (lactate), 50 mg L-theanine, 40 mg passion flower extract

3 capsules at bedtime
may cause drowsiness

Calm busy nerves
magnesium, taurine, B6, L-theanine



Mag Calm Pro
About 60 servings · Raspberry Lemon

- Supports relaxed muscles and nerves
- Supports restful sleep

IN EACH SERVING
Per scoop: 200 mg magnesium (dimagnesium malate), 500 mg taurine, 2 g myo-inositol, 100 mg L-theanine (Suntheanine), 5 mg vitamin B6 (P5P)

1 scoop in 6 oz water
in the evening

Balance calcium
vitamin K2 as MK-7



Vitamin K2
30 capsules

- Helps steer calcium into bone
- Long-acting MK-7 form

IN EACH SERVING
100 mcg vitamin K2 as menaquinone-7 (MK-7) per capsule

1 capsule daily
with your evening meal

On a statin?
CoQ10 for muscle energy



CoQ10 Ubiquinol
30 softgels

- Fuel for energy-making in muscle cells
- Statins lower the body's own CoQ10

IN EACH SERVING
100 mg ubiquinol, the ready-to-use form of CoQ10, per softgel

1 softgel daily
with breakfast

On a water pill?
magnesium + potassium



Mag-K
90 capsules

- Supports normal muscle and nerve signals
- Water pills can drain both minerals

IN EACH SERVING
Per 3 capsules: 200 mg magnesium, 99 mg potassium

3 capsules with meals
in place of Mag Calm Pro

YOUR DAILY RHYTHM

Morning
CoQ10 Ubiquinol 1 softgel

Evening meal
Mag Calm Pro 1 scoop
Vitamin K2 1 capsule

Bedtime
Muscle Eze 3 capsules

The full plan is **5 pills and 1 scoop a day**. Overwhelmed? Start with the **Start here** pick, then add one product a week.

PHARMACIST CHECK

- **Warfarin (Coumadin)?** Vitamin K2 and CoQ10 can change how it works. Start neither without your prescriber.
- **Kidney disease, spironolactone, an ACE inhibitor or an ARB?** Ask your pharmacist before Mag-K or extra magnesium. Water pills such as HCTZ can drain both.
- **Keep magnesium 4 hours away** from thyroid pills, alendronate, and antibiotics such as doxycycline or ciprofloxacin.

Special considerations for you:

Recommended by _____ Phone _____

◆ These statements have not been evaluated by the Food and Drug Administration. These products are not intended to diagnose, treat, cure, or prevent any disease. Nutrient support works alongside, never instead of, the care and medicines your practitioner prescribes.

NutriDyn